(ii) The resistance of a metal wire at 20°C is $1 \cdot 05\ \Omega$ and at 100°C is $1 \cdot 38\ \Omega$. Determine the temperature coefficient of resistivity of this metal. CBSE 26

Alternatively

$$R_1=R_0(1+\alpha\ \Delta T_1) \ldots\ldots\ldots\ldots\ldots\ldots\text{(i)}$$

$$R_2=R_0(1+\alpha\ \Delta T_2)\ldots\ldots\ldots\ldots\ldots\ldots\text{(ii)}$$

$$R_1/R_2=\frac{(1+\alpha\ \Delta T_1)}{(1+\alpha\ \Delta T_2)}$$

$$\frac{1.05}{1.38}=\frac{1+20\alpha}{1+100\alpha}$$

On solving

$$\alpha=\frac{11}{2580}\ {}^{\circ}C^{-1}$$

$$=0.0042\ {}^{\circ}C^{-1}$$

8. The resistivity ρ of a metal increases with rise in temperature because :

(A) only relaxation time 'τ' of electrons decreases with temperature.

(B) only number of electrons per unit volume 'n' increases appreciably.

(C) 'τ' decreases with temperature but 'n' does not change appreciably.

(D) 'τ' decreases with temperature and 'n' increases. CBSE 26

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(C) τ decreases with temperature but 'n' does not change appreciably.

10. Which of the following statements is true about mobility of charge carriers in a metal ?

(A) Mobility increases with increase in applied electric field.

(B) Mobility decreases with increase in temperature.

(C) Mobility is independent of the mass of the charge carrier.

(D) Mobility increases with increase in temperature

CBSE 26

10. Which of the following statements is true about mobility of charge carriers in a metal ?

(A) Mobility increases with increase in applied electric field.

(B) Mobility decreases with increase in temperature.

(C) Mobility is independent of the mass of the charge carrier.

(D) Mobility increases with increase in temperature

CBSE 26

(B) Mobility decreases with increase in temperature.

20. A heating element using nichrome is connected to a 220 V supply. Initially it draws a current of 2.9 A. After some time, the current attains a steady value of 2.5 A. Find the steady temperature of the heating element if the room temperature is 27 °C. The temperature coefficient of resistance of nichrome is $1.7 \times 10^{-4}\ ^{\circ}C^{-1}$. CBSE 26 **2**

20. A heating element using nichrome is connected to a 220 V supply. Initially it draws a current of 2.9 A. After some time, the current attains a steady value of 2.5 A. Find the steady temperature of the heating element if the room temperature is 27 °C. The temperature coefficient of resistance of nichrome is $1.7 \times 10^{-4}\ °C^{-1}$. CBSE 26 2

$R_1 = \frac{V}{I_1} = \frac{220}{2.9}\Omega$ at temperature t_1	1/2
$R_2 = \frac{V}{I_2} = \frac{220}{2.5}\Omega$ at temperature t_2	1/2
$R_2 = R_1[1+\alpha(t_2 - t_1)]$	1/2
$t_2 - t_1 = \frac{R_2 - R_1}{R_1.\alpha}$	
$t_2 - 27 = \frac{\left(\frac{220}{2.5} - \frac{220}{2.9}\right)}{\left(\frac{220}{2.9} \times 1.7 \times 10^{-4}\right)}$	
$t_2 \approx 968^0C$	1/2

18. Calculate the temperature at which the resistance of a conductor becomes 20% more than its resistance at 27 °C. The value of the temperature coefficient of resistance of the material of conductor is 2.0×10^{-4} $°C^{-1}$. **2**

CBSE 26

18. Calculate the temperature at which the resistance of a conductor becomes 20% more than its resistance at 27 °C. The value of the temperature coefficient of resistance of the material of conductor is $2.0 \times 10^{-4}\ °C^{-1}$. **2**

CBSE 26

$R_2 = R_1[1+\alpha(t_2 - t_1)]$	$1/2$
$1.2R_1 = R_1[1 + 2.0 \times 10^{-4}(t_2 - 27)]$	$1/2$
$0.2 = [2.0 \times 10^{-4}(t_2 - 27)]$	$1/2$
$t_2 = 1027^0C$	$1/2$

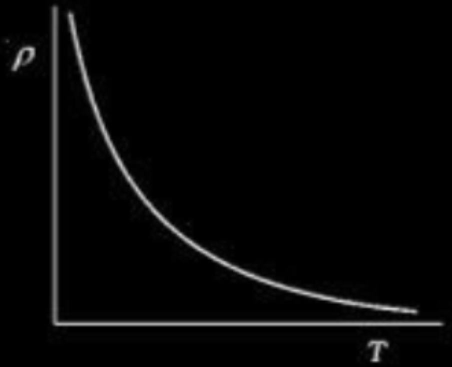
20. (a) Why manganin is used in wire-wound standard resistors ? Calculate the temperature at which the resistance of a wire increases by 25% of its resistance at room temperature 27.5 °C ? The temperature coefficient of resistance of the material of the wire is $0.004\ °C^{-1}$. **2**

(b) How does the resistivity of a semiconductor depend on temperature ? Show it on a plot.

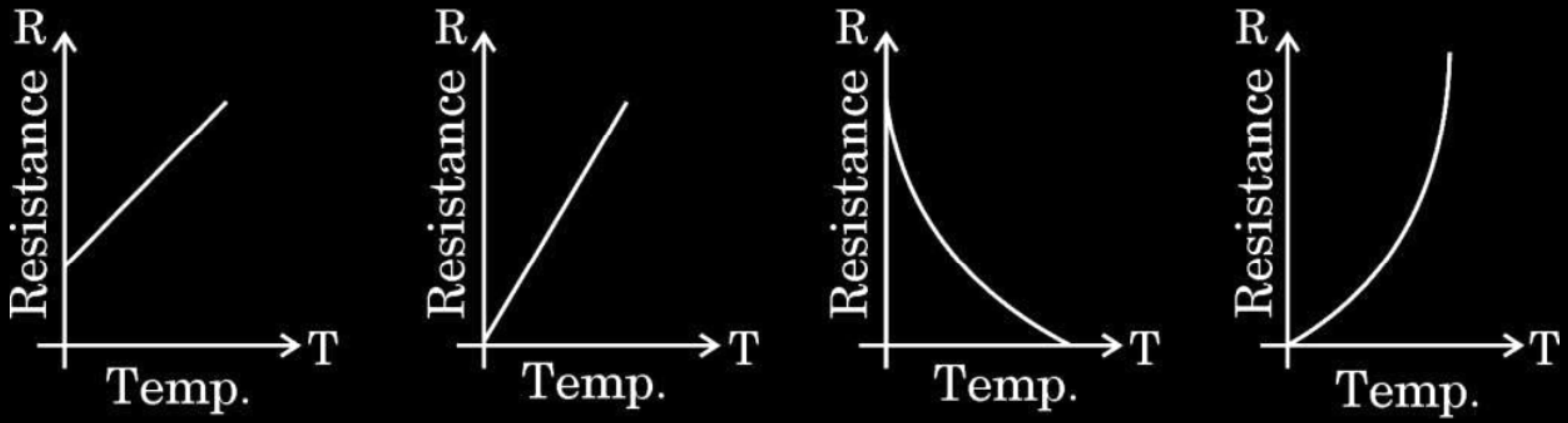
20. (a) Why manganin is used in wire-wound standard resistors ? Calculate the temperature at which the resistance of a wire increases by 25% of its resistance at room temperature 27.5 °C ? The temperature coefficient of resistance of the material of the wire is 0.004 $°C^{-1}$. **2**

(b) How does the resistivity of a semiconductor depend on temperature ? Show it on a plot.

CBSE 26

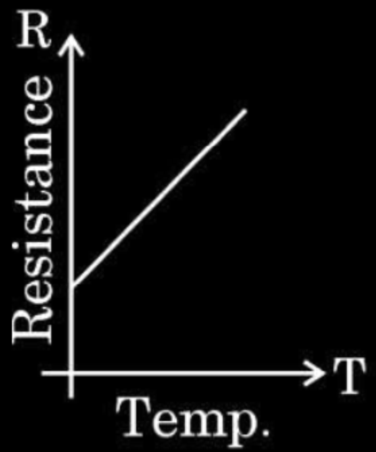
(a)	• Manganin (an alloy) exhibit a very weak dependence of resistivity with temperature $R_2 = R_1[1+\propto(t_2 - t_1)]$ $1.25R_1 = R_1[1 + 0.004(t_2 - 27.5)]$ $t_2 = 90^0C$	½ ½
(b)	• The resistivity of semiconductors decreases with increasing temperature. • 	½ ½

2. For a metallic conductor, the correct representation of variation of resistance R with temperature T is :

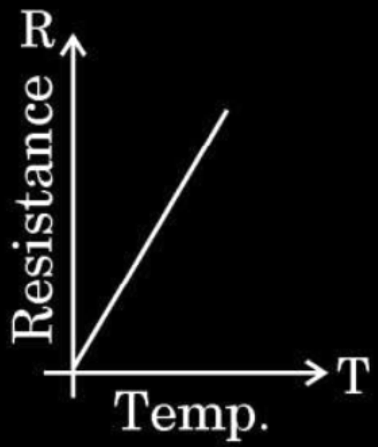


(a) (b) (c) (d)

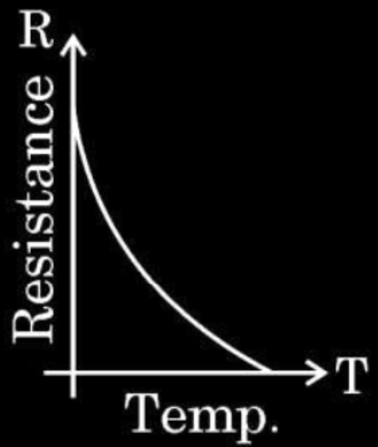
2. For a metallic conductor, the correct representation of variation of resistance R with temperature T is :



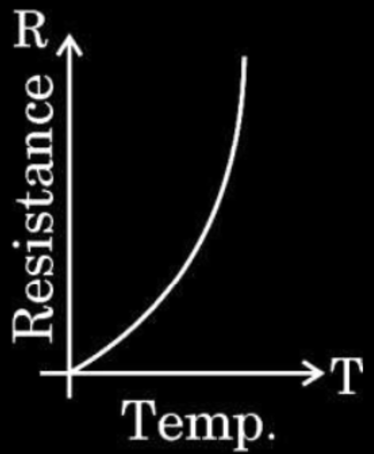
(a)



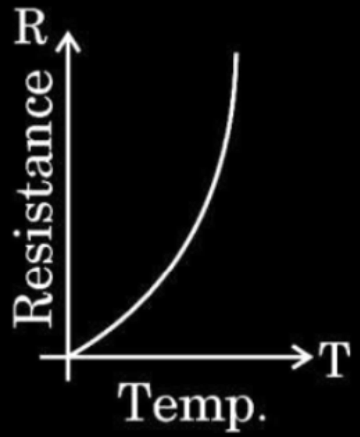
(b)



(c)



(d)



(d)

14. Pieces of copper and of silicon are initially at room temperature. Both are heated to temperature T. The conductivity of **1**

(a) both increases.

(b) both decreases.

(c) copper increases and silicon decreases.

(d) copper decreases and silicon increases.

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(d)

Copper decreases and silicon increases

16. *Assertion (A) :* The temperature coefficient of resistance is positive for metals and negative for semi-conductors.

Reason (R) : The charge carriers in metals are negatively charged whereas in semiconductors they are positively charged.

16. *Assertion (A) :* The temperature coefficient of resistance is positive for metals and negative for semi-conductors.

Reason (R) : The charge carriers in metals are negatively charged whereas in semiconductors they are positively charged.

(c)Assertion (A) is true, but Reason (R) is false.

24. (i) Define 'temperature coefficient of resistance' of a metal.

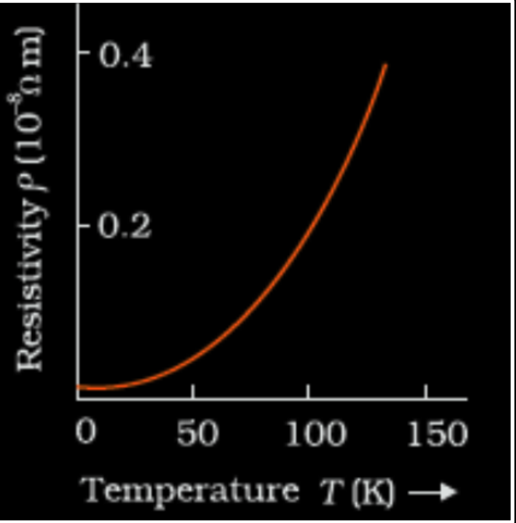
(ii) Show the variation of resistivity of copper with rise in temperature.

(iii) The resistance of a wire is 10 Ω at 27 °C. Find its resistance at –73 °C. The temperature coefficient of resistance of the material of the wire is 1.70×10^{-4} °C^{-1}.

(i) Defining temperature coefficient	1
(ii) Showing the variation of resistivity	1
(iii) Finding the resistance	1

(i) Change in resistance per unit original resistance per degree change in temperature is temperature coefficient of resistance. 1

(ii)



(iii) $R_2 = R_1(\theta_2 - \theta_1)\alpha + R_1$ ½

$$= 10(-73-27)\times 1.70\times 10^{-4} + 10$$

$$= -0.170 + 10$$

$R_2 = 9.83\,\Omega$ ½

Alternatively

$$R_1 = R_0(1+\alpha t_1)$$

$$R_2 = R_0(1+\alpha t_2)$$

$$\frac{R_1}{R_2} = \frac{(1+\alpha t_1)}{(1+\alpha t_2)}$$

$$R_2 = \frac{(1+\alpha t_1)}{(1+\alpha t_2)} R_1$$

$R_2 = \left[\dfrac{1+1.70\times 10^{-4}\times(-73)}{1+1.70\times 10^{-4}\times 27}\right]\times 10$ ½

$$R_2 = \frac{0.98759}{1.00459}\times 10\ \Omega$$

$R_2 = 9.83\ \Omega$ ½

20. Find the temperature at which the resistance of a conductor increases by 25% of its value at 27 °C. The temperature coefficient of resistance of the conductor is 2.0×10^{-4} °C^{-1}. **2**

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$R_2 = R_1 + 25\%\ \text{of}\ R_1 = 1.25R_1$	½
Temperature coefficient of resistance	
$\alpha = \dfrac{R_2 - R_1}{R_1 . \Delta T}$	½
$T_2 - 27 = \dfrac{1.25R_1 - R_1}{R_1 \times 2 \times 10^{-4}}$	½
$T_2 = 1277\ ^0C$	½

17. Find the temperature at which the resistance of a wire made of silver will be twice its resistance at 20°C. Take 20°C as the reference temperature and temperature coefficient of resistance of silver at 20°C = $4{\cdot}0 \times 10^{-3}\ K^{-1}$. *2*

17. Find the temperature at which the resistance of a wire made of silver will be twice its resistance at 20°C. Take 20°C as the reference temperature and temperature coefficient of resistance of silver at 20°C = $4{\cdot}0 \times 10^{-3}\ \mathrm{K}^{-1}$. *2*

Finding the temperature 2	
$R = R_\circ\left[1+\alpha\left(T-T_\circ\right)\right]$	½
$R = 2R_\circ$ [Given]	
$2R_\circ = R_\circ\left[1+\alpha\left(T-T_\circ\right)\right]$	½
On solving $T = T_\circ + 250$	
$T = 270°C$ or $543\ \mathrm{K}$	1

22. (a) Define the terms – 'temperature coefficient of resistance' and 'electrical conductivity' of a conductor. Why are constantan and manganin used for making standard resistances ? Explain. *3*

Definition of (i) Temperature coefficient of resistance	1
(ii) Electrical Conductivity	1
Explanation of choosing constantan and manganin for making standard resistances.	1

Temperature coefficient of resistance is defined as the fractional change in resistance per unit increase in temperature.	1
Electrical conductivity is defined as current density per unit electric field. Alternatively: Electrical conductivity is the reciprocal of electrical resistivity.	1
<u>Reason</u>: Alloys have high resistivity Low temperature coefficient of resistance Ability to achieve more resistance value in small size (Any Two)	½+ ½

3. A steady current I is passed through a conductor at room temperature for time t. It is observed that its temperature rises by 0·5°C. If 2I current is passed through the conductor (at room temperature) for the same duration, the rise in its temperature will be approximately :

(A) 1·0°C

(B) 1·5°C

(C) 2·0°C

(D) 4·0°C

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(A) 1·0°C (B) 1·5°C

(C) 2·0°C (D) 4·0°C

(C) 2.0 ^{0}C

(ii) The temperature coefficient of resistance of nichrome is $1{\cdot}70 \times 10^{-4}\ {}^{\circ}C^{-1}$. In order to increase resistance of a nichrome wire by $8{\cdot}5\%$, the temperature of the wire should be increased by : *1*

(A) 250°C

(B) 500°C

(C) 850°C

(D) 1000°C

(ii) The temperature coefficient of resistance of nichrome is $1 \cdot 70 \times 10^{-4}\ °C^{-1}$. In order to increase resistance of a nichrome wire by $8 \cdot 5\%$, the temperature of the wire should be increased by : *1*

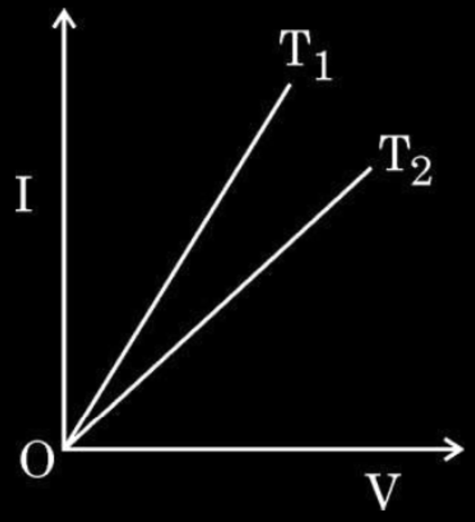
(A) 250°C

$\boxed{\text{(B)}\quad 500°C}$

(C) 850°C

(D) 1000°C

1. The figure shows the voltage (V) versus the current (I) graphs for a wire at two temperatures T_1 and T_2. One can conclude that :



(A) $T_2 = 2T_1$

(B) $T_1 > T_2$

(C) $T_1 = T_2 / 3$

(D) $T_1 < T_2$

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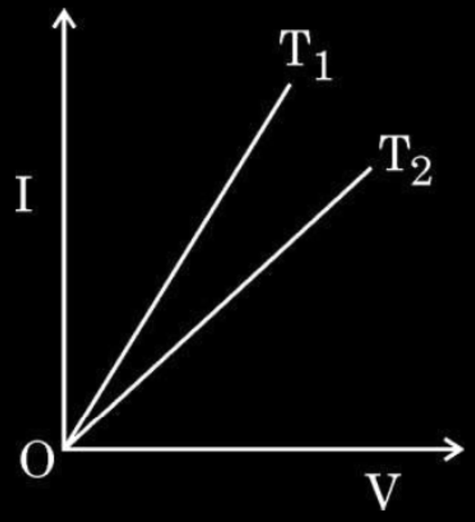

(A) $T_2 = 2T_1$

(B) $T_1 > T_2$

(C) $T_1 = T_2 / 3$

(D) $\boxed{T_1 < T_2}$

2. The resistance of a wire at 20 °C is 50 Ω. When it is heated to 120 °C, the resistance becomes 51 Ω. The temperature coefficient of resistance of the material of the wire is **1**

(A) $2 \times 10^{-4}\ °C^{-1}$

(B) $1 \times 10^{-4}\ °C^{-1}$

(C) $5 \times 10^{-4}\ °C^{-1}$

(D) $1 \times 10^{-3}\ °C^{-1}$

2. The resistance of a wire at 20 °C is 50 Ω. When it is heated to 120 °C, the resistance becomes 51 Ω. The temperature coefficient of resistance of the material of the wire is **1**

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(B) $1 \times 10^{-4}\ °C^{-1}$

(C) $5 \times 10^{-4}\ °C^{-1}$

(D) $1 \times 10^{-3}\ °C^{-1}$

21. The resistance of a wire at 25°C is 10·0 Ω. When heated to 125°C, its resistance becomes 10·5 Ω. Find (i) the temperature coefficient of resistance of the wire, and (ii) the resistance of the wire at 425°C. *2*

21. The resistance of a wire at 25°C is 10·0 Ω. When heated to 125°C, its resistance becomes 10·5 Ω. Find (i) the temperature coefficient of resistance of the wire, and (ii) the resistance of the wire at 425°C. *2*

Finding (i) temperature coefficient of resistance 1 ½ (ii) resistance of wire at 425 °C ½	
(i) $R_2 = R_1(1+\alpha(t_2-t_1))$	½
$10.5 = 10(1+\alpha\times100)$	
$\alpha = 5\times10^{-4}\,/\,^{o}C$	½
(ii) $R_{425} = R_{25}(1+\alpha(425-25))$	½
$= 10(1+5\times10^{-4}\times400)$	
$= 12\,\Omega$	½

(a) Draw a graph showing the variation of current versus voltage in an electrolyte when an external resistance is also connected.

(b) (i) The graph between resistance (R) and temperature (T) for Hg is shown in the figure (a). Explain the behaviour of Hg near 4 K.

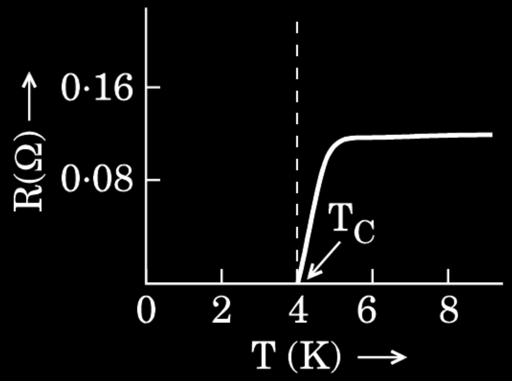


Figure (a)

(ii) In which region of the graph shown in the figure (b) is the resistance negative and why ?

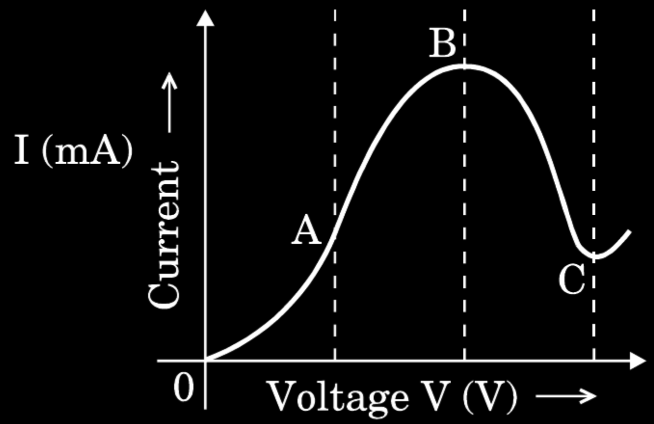
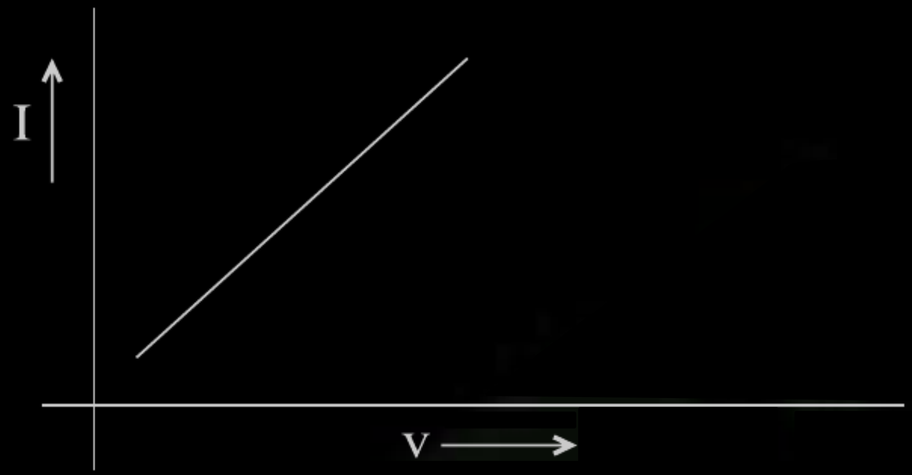
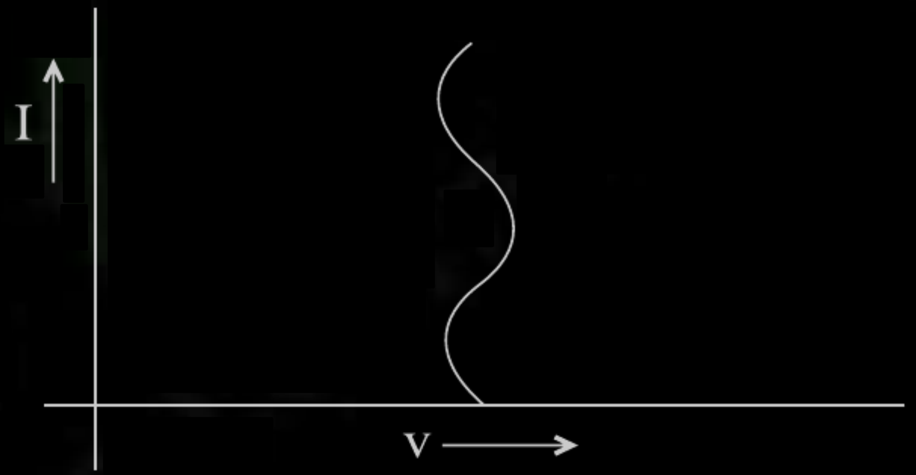


Figure (b)

a) Note : Award this 1 mark even if the student draw the following or some other non linear graph 1



OR



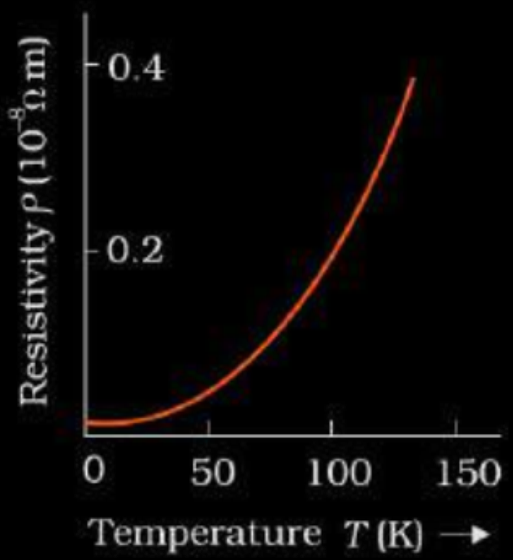
b) At a temperature of 4 K, the Resistance of Hg becomes zero. 1
Note - Award this 1 mark if the student writes that Hg becomes a super conductor at temperature of 4 K

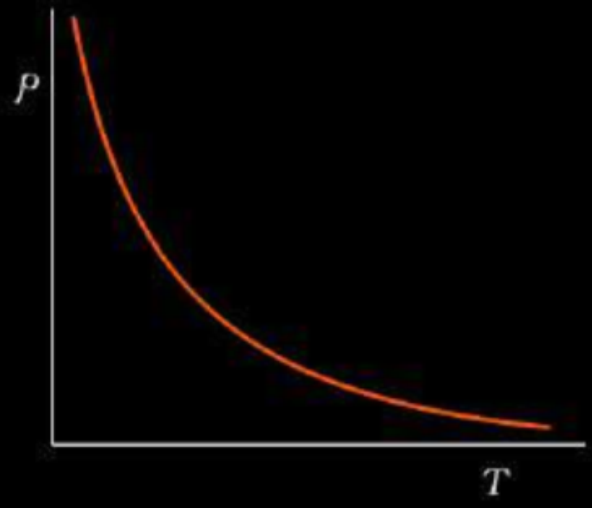
c) Region BC ; since current is decreasing with increasing voltage 1
Alternately :The graph has negative slope for this region.

Show, on a plot, variation of resistivity of (i) a conductor, and (ii) a typical semiconductor as a function of temperature.

Using the expression for the resistivity in terms of number density and relaxation time between the collisions, explain how resistivity in the case of a conductor increases while it decreases in a semiconductor, with the rise of temperature.

• Showing the plot of variation of resistivity	½ + ½
• Expression for resistivity	1
• Explaining variation of resistivity for conductor and semiconductor	½ + ½





½

½

- $\rho = \dfrac{m}{ne^2\tau}$ 1
- In case of conductors with increase in temperature, relaxation time decreases, so resistivity increases. ½
- In case of semiconductors with increase in temperature number density (n) of free electrons increases, hence resistivity increases. ½

(i) Define the term drift velocity.

(ii) On the basis of electron drift, derive an expression for resistivity of a conductor in terms of number density of free electrons and relaxation time. On what factors does resistivity of a conductor depend ?

(iii) Why alloys like constantan and manganin are used for making standard resistors ?

i) Average velocity acquired by the electrons in the conductor in the presence of external electric field.	1
[Alternatively: $v_d = \frac{-eE\tau}{m}$ where τ is the relaxation time.]	
ii) $v_d = \frac{-eE\tau}{m}$	
We have $E = -\frac{V}{\ell}$, where V is potential difference across the length 'ℓ'of the conductor	
$v_d = \frac{eV\tau}{m\ell}$	½
Current flowing $I = neAv_d$	½
$I = neAv_d \frac{eV\tau}{m\ell} = \frac{ne^2AV\tau}{m\ell}$	½
$\frac{I}{V} = \frac{ne^2A\tau}{m\ell} = \frac{1}{R}$ -----------(i)	
Also, R = $\rho \frac{\ell}{A}$ -----------(ii)	
Comparing (i) and (ii)	
$$\rho = \frac{m}{ne^2\tau}$$	½
Resistivity of the material of a conductor depends on the relaxation time, i.e., temperature and the number density of electrons.	½+ ½
iii) Because constantan and manganin show very weak dependence of resistivity on temperature	1